Exponent study, extension: m = 3 and m = 4 at larger |V| under

raised limits (2026-11-12) **Class.** An engine measurement in counted units: F_p multiplications of the F4 row reduction, `F4Report.field_ops`, from `curve25519_family f4sub`, with the same frozen binary, seeds and cell format as the parent study `../curve25519_exponent_study_20261108/`. It measures the cost T(E,m,L) of one symmetrised decomposition system against the base size |V|. It is not an end-to-end solve, not an ecbench session, and not a leaderboard or `benchmarks/` row. No pcc claim was registered, so every measured number below is an unverified assertion. Rho remains ahead on every track [pcc:unverified].

Question (owner, 2026-11-10). The parent study left the m = 3 and m = 4 verdicts "undetermined" because its limits (3.5 GB, 1,200 s per target) censored the series before the preregistered estimator could apply. This extension raises the limits to 12 GB of virtual memory per process and 10,800 s per target, adds the even divisors of p - 1 at k = 20 (the parent used odd d only), and reruns the parent's censored cells. The estimator and the lever thresholds are the parent's (section 5 there): a lever at m = 4 needs the fitted exponent s of T in |V| below 0.5 with upper 95 % bound below 0.5; at m = 3 it needs s below 0 with upper bound below 0.

Answer. The verdicts at m = 3 and m = 4 stay "undetermined" under the preregistered estimator, and no reading meets a lever rule [pcc:unverified]. Four new cells completed, all at k = 20, m = 3 (d = 12, 14, 18, 28). The four larger cells attempted (k = 20, m = 3, d = 36; k = 20, m = 4, d = 12 and 19; k = 22, m = 3, d = 55) exhausted 12 GB of memory; none was censored by time [pcc:unverified]. The engine writes its record only at the end of a cell, so how many targets had finished at the abort is not recorded. The binding limit on this route at these sizes is memory, by inference the F4 matrix, not wall time.

- **Preregistered estimator: still not applicable.** It needs at least 4 complete cells with d >= 9 spanning 4x in |V|. At k = 20, m = 3 there are now 8 such cells (d = 9 to 28) but they span 3.1x; the span gate needed d = 36, which censored. At m = 4 only d = 9 is complete above the boundary cells. At k = 22, m = 3 nothing changed (6 cells, 3.7x) [pcc:unverified].
- **Other readings (not verdicts).** The d >= 9 fit at k = 20, m = 3, computed despite the span gate, is now s = 6.08 [1.69, 10.47] over 8 cells; in the parent it was 5.56 [−8.75, 19.88] over 4 cells. The extension cells therefore move the lower bound of this reading above 0, so the d >= 9 cells alone now exclude s <= 0 at 95 % at m = 3, k = 20 [pcc:unverified]. The sensitivity fit over all 10 complete cells (d = 3 to 28) gives 6.87 [5.01, 8.73]; the four even cells alone give 7.15 [1.38, 12.92]; the odd cells with d >= 9 alone give 5.56 [−8.75, 19.88] [pcc:unverified]. Every point estimate is far above both thresholds (0 and 0.5), and no upper bound is below them.
- **Cost does not follow |V| alone.** Three of the four even cells are much dearer than their odd neighbours: d = 18 cost 2.43e10 field operations against 2.20e8 at d = 19 (110x), and d = 12 cost 7.75e8 against 1.48e8 at d = 9 (5x), while d = 14 cost the same as d = 12 and d = 28 cost 0.77x of d = 27 [pcc:unverified]. The validator's exploratory joint fit of ln T on ln d and on the number of prime factors of d (10 cells, k = 20, m = 3) gives 5.16 [4.15, 6.16] in ln d and 2.14 [1.35, 2.93] per prime factor; it is exploratory, not preregistered [pcc:unverified]. The symmetrised system has one chain variable per prime factor of d per summand (d = 19: 1; d = 14: 2; d = 12, 18, 28: 3), so the number of prime factors of d, with multiplicity, matters as much as |V| = d. A single power law in |V| is a coarse model for this series, which is one more reason the span gate was not met.

1. Predictions (frozen in `PROTOCOL.md` before

data) and outcomes

Prediction	Outcome	Basis
P1: k = 20, m = 3, d = 57 censors again at 10,800 s	**not testable**	the chain stopped at d = 36 (memory), so d = 57 was never rerun under the chain rule
P2: >= 4 even m = 3 cells complete, the estimator applies at k = 20, m = 3, s > 3 with lower bound > 0	**failed**	4 even cells completed (12, 14, 18, 28), but the estimator does not apply: 8 cells with d >= 9 span 3.1x < 4x
P3: k = 20, m = 4, d = 19 censors under 12 GB / 10,800 s	observed, but outside the chain rule (see below)	memory allocation failure after 1,027 s; the cell was run after d = 12 had censored, which the protocol's chain rule on the union of cells excludes
P4: m = 4 cells d = 12, 14, 18 complete, all two-point slopes above 2	**failed**	d = 12 exhausted 12 GB after 6,226 s; d = 14 and 18 were not attempted (chain rule)
P5: k = 22, m = 3, d = 55 censors again (memory)	passed	memory allocation failure after 5,286 s
P6: binary sha256 matches `data/frozen.sha256` at launch; rerun cells' first target_x equals the parent's (610025 for d = 57)	passed in part	the hash matched at each launch and in every cell record; the d = 57 part is not testable (not rerun); the rerun cells d = 19 (m = 4) and d = 55 (k = 22) aborted before the engine recorded a target, so no target_x comparison exists

Chain-rule deviation (validator erratum E1). The protocol applies the chain rule to the union of parent and extension cells in d order, under which the k = 20, m = 4 chain ends at d = 12. The frozen runner kept the lists 12, 14, 18 and 19, 21, ... as

separate chains, so it ran d = 19 for 1,027 s after d = 12 had censored (`data/run.log` prints "censored at d=12" and then runs d = 19). Under the protocol's rule P3 is therefore not testable, like P1; the d = 19 record is kept as an observation (it censored by memory, as P3 said) and is not counted as a passed prediction. The deviation is in the frozen runner, not a post-hoc choice, and the cell affects no fit.

Failed predictions stay failed. P2 and P4 both failed on the resource side: the memory limit, not the measured slopes, decided them. The slope part of P2 (s > 3, lower bound > 0) holds in every reading that has a lower bound above 0, but that is not the preregistered estimator.

2. Cells

Seeds are the parent's (20261108 + 1000 m + d), 6 targets per cell, G = 0, form `sym`, one process at a time with `RAYON_NUM_THREADS=1`. T is the mean of `field_ops` over the 6 targets; the six values agree to within 1e-5 relative in every complete cell. All 24 completed targets were inconsistent systems (a uniform target is a sum of three base points of a base this small with negligible probability), as in the parent.

k	m	d	prime factors	|V|	T (field ops)	solving degree	max columns	wall	outcome
20	3	12	2·2·3	12	7.748e8	7	7,759	16 s	complete
20	3	14	2·7	14	7.766e8	10	10,651	18 s	complete
20	3	18	2·3·3	18	2.4275e10	7	13,616	298 s	complete
20	3	28	2·2·7	28	2.0161e11	11	54,949	3,693 s	complete
20	3	36	2·2·3·3	36				4,521 s	censored: memory, 12 GB
20	4	12	2·2·3	12				6,226 s	censored: memory, 12 GB
20	4	19	19	19				1,027 s	censored: memory, 12 GB (parent: memory, 3.5 GB)
22	3	55	5·11	55				5,286 s	censored: memory, 12 GB (parent: memory, 3.5 GB)

Every censor is an `exit -6` with "memory allocation of N bytes failed" on stderr under `ulimit -v 12000000`; no target reached the 10,800 s cap. Cells d = 38 to 133 at k = 20, m = 3, d = 21 to 57 at k = 20, m = 4, and d = 75, 93 at k = 22, m = 3 were not attempted, by the chain rule.

Parent cells used in the fits (unchanged): k = 20, m = 3: d = 3, 7, 9, 19, 21, 27; k = 20, m = 4: d = 3, 7, 9; k = 22, m = 3: d = 3, 5, 11, 15, 25, 31, 33, 41. Full numbers: `data/analysis.json`.

3. Fits

The estimator is the parent's section 5: OLS of ln T on ln |V| over complete cells with d >= 9, applied only with at least 4 cells spanning 4x, two-sided 95 % t-interval. `analysis.json` uses the parent's 3-decimal t table; the intervals quoted here use exact quantiles (validator erratum E2), which moves two bounds by 0.01.

group	preregistered fit (d >= 9, gate)	d >= 9 despite the gate	all complete cells (sensitivity)
k = 20, m = 3	not applicable: 8 cells, span 3.1x	6.08 [1.69, 10.47], 8 cells	6.87 [5.01, 8.73], 10 cells, span 9.3x
k = 20, m = 4	not applicable: 1 cell	(fewer than 3 cells)	not applicable: 3 cells, span 3.0x
k = 22, m = 3	not applicable: 6 cells, span 3.7x	4.51 [−1.27, 10.29], 6 cells	5.76 [3.77, 7.75], 8 cells, span 13.7x

Parent values for comparison: k = 20, m = 3 d >= 9 despite the gate 5.56 [−8.75, 19.88] (4 cells); all cells 6.44 [3.49, 9.39] (6 cells). The k = 22 row is the parent's, since d = 55 censored again.

Two-point local slopes at k = 20, m = 3 in d order: 9 to 12: 5.8; 12 to 14: 0.0; 14 to 18: 13.7; 18 to 19: −87.0; 19 to 21: 25.4; 21 to 27: 18.1; 27 to 28: −7.4. The two negative slopes are the two steps from an even d to the next odd d.

Under the paper's model (omega = 2) the relation and linear-algebra phases balance at total exponent 2/(m + 1 - s). At the point estimates (s >= 4.5 in every reading) the denominator is at most −0.5 at m = 3 and at most 0.5 at m = 4, so no balance below the generic 1/2 exists at either m. This rests on point estimates: the d >= 9 intervals at k = 22 (m = 3) and the only m = 4 fit, over the three cells d = 3, 7, 9, 9.86 [−59.9, 79.6], still include s < 0. The extrapolation is derived from a fixed small p, not measured.

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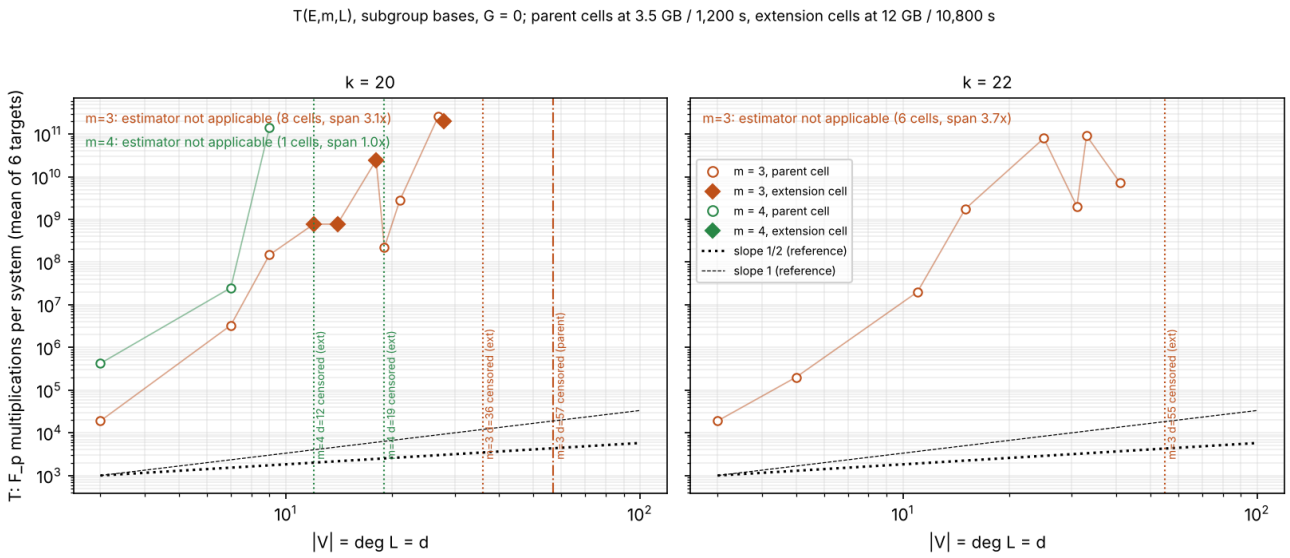


Figure 1: T(E,m,L) against |V| at m = 3 (orange) and m = 4 (green), k = 20 (left) and k = 22 (right); open circles parent cells, filled diamonds extension cells, vertical lines censored cells. Every number is an unverified assertion.

4. Protocol, freeze and changes after the freeze

- `PROTOCOL.md` and `scripts/run_ext.py` were frozen in commit `0afe01c59d` (2026-11-10T15:38Z), before any cell. `data/frozen.sha256` is the parent's file (protocol, parent scripts and binary hashes); the binary line is the one this study checks.
- Post-freeze fix before any data (`4424e037d8`, 15:39Z): the runner's launch check read the first line of `frozen.sha256` (the parent protocol's hash) instead of the binary's line and refused to start. The fix selects the line ending in `curve25519_family`. No cell had run.
- `scripts/analyse_ext.py` and `scripts/figure.py` were written after the freeze, while the last chain was running (7 of 8 cells done). The "other readings" block of `analyse_ext.py` (fit despite the gate, even/odd split) was added in `723609651a`, after the last cell was committed; the diff adds readings only and touches no verdict or preregistered number. `analyse_ext.py` reimplements the parent's frozen estimator on the union of parent and extension cells and encodes the P1 to P6 rules of the protocol; its "other readings" (fit despite the gate, even/odd split) are not preregistered and are labelled so above. The union rule: an extension cell replaces the parent cell with the same key and d; the parent's censored d = 57 was not rerun and is kept as the parent's.
- A smoke test of an even divisor (d = 4, m = 2, 1 target, about 30 s) ran before the freeze to check that the engine accepts even d; it is not a study cell.

5. Host, restarts and other disclosures

- **Host.** A shared 4-core, 15 GB container of the same class as the parent's; no EKS, Conductor, S3 checkpoint or benchmark record (`benchrec`) because none of them is reachable from this runtime; this is a deviation from the repository's default execution path, disclosed as in the parent. Load averages (1 min) before and after each cell are in each cell's `ext` block; they were 1.0 to 1.5 for most cells, 3.0 before d = 18, 2.5 before d = 12 at m = 4 and 4.6 before k = 22 d = 55. Wall time is not a claimed quantity; only `field_ops` is, and memory censoring does not depend on load.
- **Three container restarts** killed the detached runner (`data/env.txt`). The times in `env.txt` are when a fresh container was found, not when the runner died: the first kill was around 15:40 to 15:45Z on 2026-11-10 during d = 18 (found 19:40Z), the second on 2026-11-10 during d = 12 at m = 4 (found 20:12Z on 2026-11-11, about 23 h later), and the third around 01:43Z on 2026-11-12 during d = 55 at k = 22 (relaunched 01:44Z). The kill windows are inferred from the cells' wall times. The kernel changed from 6.18.44-fc-v338 to v378 across the run, so "the same container" means the same environment class and limits, not one machine. The runner writes a cell file only when the cell ends, so no partial data exists; each interrupted cell was rerun from scratch with the same seed. The second check-in trigger reached the session about 20 hours late, which is why the run spanned two days. After the second restart a tracked background waiter kept the session active between cells.
- **Binary.** `/home/user/bin-archive/curve25519_family-dc2c30a4`, sha256 `dc2c30a4242d80281622b2cb42cbaf9cb457d041ffbc9a4d35794bb893bf3faf`, equal to the parent's frozen hash at every launch (`data/run.log`, "binary ok") and recorded in every cell's `ext.bin_sha256`. It is not committed to Git; the parent's artifact-store upload covers its raw records, not the binary.
- **Memory accounting.** `ulimit -v 12000000` limits virtual address space, so the engine may have failed below 12 GB of resident memory. The failure point is therefore an upper bound on what the matrix needs at that cell, not a measurement of it.

6. What this changes and what is next

- The parent's open item "the m = 3 and m = 4 exponent cells need more memory and time than the local container allows" is now quantified: 12 GB does not reach the span gate at m = 3 (d = 36 fails) and does not reach a second non-boundary cell at m = 4 (d = 12 fails). Time was never the limit at these sizes.
- Applying the preregistered estimator at m = 3 needs one of: more memory for d = 36 (k = 20), or a host where k = 22 d = 55 fits; at m = 4 it needs d = 36 or beyond, which is out of reach of this engine's dense F4 at any plausible memory. The alternative is to change the engine (sparse elimination, or a Gröbner engine with a degree bound), which is an engine change, not a measurement, and would need its own freeze.
- Nothing here is an improvement over the index-calculus baseline, and nothing is closed: the exponent question at m = 3 and m = 4 stays undetermined. No gap-to-rho figure is reported, because T is the cost of one decomposition system, not of a solve; the parent's section 7 gives the derived comparison [pcc:unverified].

7. Files

File	Content
`PROTOCOL.md`	frozen protocol (cells, limits, estimator, predictions P1 to P6)
`scripts/run_ext.py`	frozen sequential, resumable runner
`scripts/analyse_ext.py`, `data/analysis.json`	estimator on the cell union, prediction decisions, other readings
`scripts/figure.py`, `exponent_ext.svg`, `exponent_ext.png`	figure: T against
`cells/*.json`	the 8 cell records (parent format plus an `ext` block with limits, load, wall, binary hash, exit)
`data/run.log`, `data/env.txt`, `data/frozen.sha256`	run log, launch and restart notes, frozen hashes
`REPORT.typ`, `REPORT.pdf`	the report (pandoc to Typst, compiled with the typst Python module)
`validator-report.json`	independent validation (see "Validation")

Validation

Independent validation on 2026-11-12 of commit `236e925fb7` (`validator-report.json`, scripts in `validation/`): status `pass_with_errata`, disposition `validated`, `independent: true`. The validator reimplemented the estimator, the curve, the targets and the consistency test without the producer's scripts; regenerated all 24 recorded targets and the parent's d = 57 first target from the seeds; confirmed with its own S_3 and S_4 that all 24 systems are inconsistent; re-serialised every cell file byte-identically with the runner's writer; checked the freeze order, the binary hash, the timeline and every README number; and replayed d = 12 and d = 14 with the frozen binary, matching every field except per-target milliseconds. Of 74 headline values, 72 agreed and 2 differed by 0.01 (E2). Errata E1 to E11 are applied above: E1 chain-rule deviation at m = 4 (P3 rescored), E2 t-quantile rounding, E3 restart times, E4 a missing load average, E5 kernel change, E6 claims the records cannot show, E7 parity wording, E8 post-data script edit, E9 rho and closure wording, E10 the balanced-total paragraph, E11 figure cosmetics. Not done by the validator: replay of d = 18 and d = 28 (their targets and flags were checked, not their field_ops), rerun of the censored cells, rebuild of the binary from source, and the session-timing statements. The numbers above remain [pcc:unverified] under the repository's claim rule because no pcc claim was registered; the validator report is the evidence a claim would cite.